

Matthew Ju

matthewju.com | matthew.ju@outlook.com | github.com/matthewju06 | linkedin.com/in/matthew-ju

Education

University of Illinois Urbana-Champaign

Bachelor of Science in Computer Science & Statistics — GPA: 3.80/4.00

Expected May 2027

Champaign, Illinois

Projects

Agent-Assisted Grocery Planning System

Python, PostgreSQL, FastAPI, SQL, OpenAI API, Instacart Developer Platform API

- Building a backend-first decision support system to optimize grocery shopping under dietary, caloric, and budget constraints
- Designing relational data models for users, recipes, ingredients, and substitutions to enable deterministic optimization over structured data
- Implementing an agent-assisted interface using structured tool calls (MCP) to translate natural language preferences into validated planning constraints
- Integrating with Instacart's Developer Platform APIs to generate shoppable grocery lists and hand off checkout to external fulfillment

ListingLab - Market Analytics Platform

Python, Flask, JavaScript, Chart.js, OAuth 2.0, REST APIs

- Built a market analytics platform ingesting up to 400 listings per query from the eBay Browse API using a Python Flask backend, enabling reproducible market searches and comparative analysis
- Optimized external API throughput with parallel requests, OAuth token caching, and Redis-based rate limiting, reducing end-to-end query latency from 4s to 1s on average
- Implemented custom analytics logic including search history tracking, price aggregation, and an adaptive IQR-based price filtering algorithm, reducing outlier inclusion by 35–45% compared to raw marketplace data
- Designed a Model Context Protocol (MCP) interface exposing structured market analytics tools, enabling LLMs to query filtered price distributions and summarized market data

Technical Experience

Corteva

Incoming Software Engineering Intern (R&D)

Summer 2026

Johnston, Iowa

AbbVie

Data Engineering Intern

June 2024 – August 2024

North Chicago, Illinois

- Built a real-time manufacturing data pipeline reading sensor data from industrial equipment over ethernet and wireless networks into Python-based processing workflows
- Implemented streaming signal processing in Python using moving average and median filters to denoise sensor signals and detect experimental anomalies in near real time
- Automated data extraction and visualization, reducing experimental processing time by 50%

Skills

Languages: Python, SQL, Java, C++

Backend & Data: FastAPI, Flask, PostgreSQL, REST APIs, OAuth 2.0, Docker, OpenAI API, eBay API, Instacart API

Frontend: React, Next.js, TypeScript, Chart.js

Additional Experience

YJ Consulting

Co-Founder

May 2025 – December 2025

Vernon Hills, Illinois

- Led client restaurant website modernization and digital menu optimization, increasing viewership by 37% and sales by 8%; built company website: <https://yjconsulting.pro/>

Champaign Urbana Business and Engineering (CUBE) Consulting

Technical Consultant

February 2025 – December 2025

Champaign, Illinois

- Conducted feasibility analysis for a student startup's SaaS platform to define technical and financial requirements
- Led website planning for an on-campus cafe